

CSCI E-106: Assignment 3

Due Date: February 20, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

Refer to the Crime rate data. A criminologist studying the relationship between level of education and crime rate in medium-sized U.S. counties collected the following data for a random sample of 84 counties; X is the percentage of individuals in the county having at least a high-school diploma, and Y is the crime rate (crimes reported per 100,000 residents) last year. (100 points)

a-) Create train data set by selecting 70% of the data randomly and use the remaining 30% of the data as a test data set (use `set.seed(10)` before selecting the sample). (5 points)

Use train data set to answer questions for part b to i

b-) Build a regression model to predict Y on the train data set. Comment on your model. Are there any outliers in the data set? Are there errors normally distributed? (25 points)

c-) Set up the ANOVA table for the regression model. (5 points)

d-) Carry out the test in part a by means of the F test. Show the numerical equivalence of the two test statistics and decision rules. Is the P -value for the F test the same as that for the t test? (5 points)

e-) By how much is the total variation in crime rate (Y) reduced when percentage of high school graduates (X) is introduced into the analysis? Is this a relatively large or small reduction? (5 points)

f-) State the full and reduced models. (5 points)

g-) Obtain (1) $SSE(F)$, (2) $SSE(R)$, (3) dfF , (4) dfR , (5) test statistic F^* for the general linear test, (6) decision rule. (5 points)

h-) Are the test statistic F^* and the decision rule for the general linear test numerically equivalent to those in part b? (5 Points)

i-) Evaluate model performance and stability on the train data set (20 points)

Use test data set below

j-) Evaluate model performance and stability on the test set and compare it to the training set. Comment on your findings? (20 points)